

Chapter

Metallurgy

1. Copper, Zinc and Tin are the metals alloyed to form: MAIN 2024

- (a) Duralumin
- (b) Brass
- (c) Bronze
- (d) Solder

2. The compound that is not a constituent of the electrolytic mixture used in the Hall-Heroult's process is: MAIN 2024

- (a) Al_2O_3
- (b) NaAlO_2
- (c) Na_3AlF_6
- (d) CaF_2

3. A mineral from which the metal can be extracted economically and conveniently is known as: MAIN 2024

- (a) Matrix
- (b) Ore
- (c) Flux
- (d) Alloy

4. The ore of aluminium is. COMP 2022

- (a) calamine
- (b) haematite
- (c) magnetite
- (d) cryolite

5. The process used to purify alumina by electrolytic reduction. MAIN 2022

6. Bonding present in metallic chlorides. SQP 2021

7. The process of removal of Gangue from ore. MAIN 2023

8. **Assertion** : Alkali metals do not form dipositive ions.

Reason : After loss of one electron alkali metals achieve stable electronic configuration of noble gases. SQP 2025

(a) Both assertion and reason are true and reason is the correct explanation of reason.

(b) Both assertion and are true but reason is not the correct explanation of assertion.

(c) Assertion is true but reason is false.

(d) Assertion is false but reason is true.

9. During termite welding _____ acts as a reducing agent. SQP 2023

10. Brass is an alloy of _____ and Zinc. MAIN 2021

11. Reaction of Zinc with hot concentrated Sodium hydroxide. COMP 2023

12. $\text{CuCO}_3 \xrightarrow{\text{Heat}}$ SQP 2021

13. $\text{ZnO} + \text{H}_2\text{SO}_4 \xrightarrow{\text{Heat}}$ COMP 2023

14. Why is aluminium used in flash bulbs ? MAIN 2023

15. How many valence electrons are present in :

(i) metals

(ii) non-metals ? COMP 2021

16. What is the special property of : MAIN 2023

(i) Duralumin

(ii) Type metal ?

17. Name the alloy which is made up of: SQP 2025

(i) Copper, Zinc and Tin

(ii) Lead and Tin

18. Name the main metal used in making of alloys given below : MAIN 2024

- (a) Duralumin
- (b) Stainless steel

19. State the relevant reason for the following :

- (i) Concentrated alkali is used for the concentration of bauxite ore.
- (ii) Fused alumina is reduced to aluminium by electrolysis. MAIN 2022

20. State one use of the given alloys : SQP 2022

- (i) Magnalium
- (ii) Duralumin

Solution

1. (d) Solder

- **Solder** is an alloy used for joining metals.
- There are **two types of solder**:
 1. **Soft solder** – an alloy of **tin (Sn)** and **lead (Pb)**, used for delicate work.
 2. **Hard solder** – an alloy of **copper (Cu)** and **zinc (Zn)**, used for stronger joints.

2. (b) NaAlO_2

- **Hall-Hérault process** is used for the extraction of **aluminium** by electrolysis.
- The **electrolytic mixture** typically contains:
 - Al_2O_3 (**aluminium oxide**) – the source of aluminium
 - CaF_2 (**calcium fluoride**) – to lower the melting point
 - Na_3AlF_6 (**cryolite**) – acts as a solvent and reduces melting point
- NaAlO_2 is **not used** in the electrolytic mixture.

3. (b) Ore

- An **ore** is a naturally occurring mineral that contains a **high concentration of a desired metal**.
- It is **economical and convenient** to extract the metal from an ore because of its rich metal content.
- Other terms:
 - **Matrix** – the surrounding rock of the ore

- **Flux** – used in metallurgy to remove impurities
- **Alloy** – a mixture of two or more metals

4. (d) Cryolite

- The **main ore of aluminium** is **bauxite**.
- Other ores include **cryolite** and **corundum**.
- These ores contain **aluminium compounds** from which aluminium can be extracted.

5. The process used to purify **alumina** (Al_2O_3) by **electrolytic reduction** is called the **Hoope's process**. This method involves dissolving alumina in molten cryolite and then electrolyzing it to obtain pure aluminium metal.

6. Ionic bonding

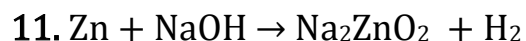
7. Concentration

8. (a) Both assertion and reason are true and reason is the correct explanation of assertion

Alkali metals have **one electron in their outermost shell**. They lose this single electron to achieve the **stable electronic configuration of noble gases**, forming **monopositive ions** (M^+). Since they achieve stability after losing just one electron, they **do not form dipositive ions** (M^{2+}). Hence, the assertion and reason are both true, and the reason correctly explains the assertion.

9. aluminium

10. copper



14. Aluminium is used in **flash bulbs** because it **burns in oxygen** with a **bright, intense white light**. This property makes aluminium ideal for producing the strong flashes of light required in photography.

15. **Metals:** 1 to 3 valence electrons in their outermost shell.

Non-metals: 4 to 7 valence electrons in their outermost shell.

16. (i) **Duralumin:** It is a special alloy that is **light in weight** yet as **strong as steel**. It is **highly ductile, hard, and resistant to corrosion**, making it ideal for use in **aircraft and structural applications**.

(ii) **Type metal:** This alloy has a **low melting point**, which allows it to **melt and cast easily**. Additionally, it **expands slightly on solidification**, making it suitable for **printing types in the printing industry**.

17. (i) Bronze

(ii) Solder

18. (a) **Duralumin:** The main metal used in duralumin is **aluminium (Al)**. It is alloyed with **copper (Cu), manganese (Mn), and magnesium (Mg)** to increase its strength and hardness.

(b) **Stainless steel:** The main metal in stainless steel is **iron (Fe)**. It is alloyed with **chromium (Cr)** to provide **corrosion resistance** and durability.

19. (i) Concentrated alkali is used for the concentration of bauxite because **bauxite ($\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$)** **dissolves only in hot concentrated alkali** such as NaOH, while the **impurities remain insoluble**. This allows the impurities to be **easily separated by filtration**.

(ii) Fused alumina is reduced to aluminium by **electrolysis** because **alumina is highly stable and cannot be reduced by ordinary reducing agents**. Electrolysis provides the necessary energy to obtain aluminium metal.

20. (i) **Magnalium:** This alloy is used in **aircraft and automobile parts** because it is **lightweight and strong**.

(ii) **Duralumin:** Duralumin is used in **pressure cookers** due to its **strength, lightness, and resistance to corrosion**.